VOLTAMMETRY

Voltammetry is the general name given to a group of electro analytical methods in which the current is measured as a function of applied potential where in the polarization of the indicator or working electrode is enhanced. The field has been developed from polarography.The word polarography first recorded in 1935 – 1940 Polaro(graph) + graphy(field of study)

Polarography

Polarography is created by **Jaroslav Heyrovsky** for that he awarded Nobel Prize in 1959.It is an electrochemical technique of analysing solutions that measure the current flowing between two electrodes in the solution as well as the gradually increasing applied voltage to determine respectively the concentration of solute and its nature.

PRINCIPLE

The main principle in the polarography is the reduction process taking place at the electrode. This method has limited sensitivity. The reduction at the electrode increases the voltage applied between the polarisable and non-polarisable electrodes and the current is recorded that is, the metallic ions are reduced at the surface of the electrode. Study of solutions or of electrode processes by means of electrolysis with two electrodes, one polarizable and one unpolarizable, the former formed by mercury regularly dropping from capillary tube.

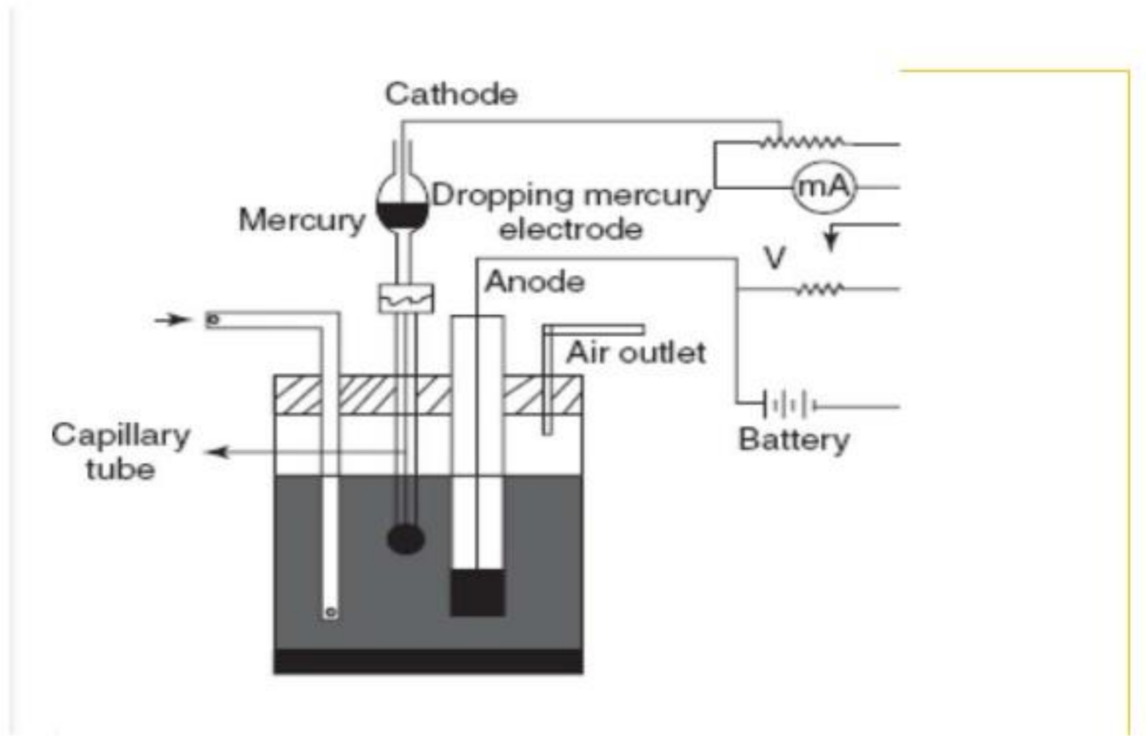
POLARIZED ELECTRODE: Dropping Mercury Electrode (DME)

DEPOLARIZED ELECTRODE: Saturated Calomel Electrode

INSTRUMENTATION

The apparatus consists of a dropping mercury electrode which acts as a cathode and as a working electrode. The anode used is the pool of mercury at the bottom of the reservoir which acts as a reference electrode. The reference electrode potential is constant. These two electrodes are placed in the sample solution which contains the both anions and cations. Then these anode and cathode are connected to the battery, voltammeter and galvanometer. Then apply the constant voltage and record the current–voltage curves using recorders.

polarography apparatus:



ELECTRODES:

The polarography is mainly composed of the three types of the electrodes. They are as follows:

WORKING ELECTRODES

The working electrode is mainly used for the determination of the analyte response to the potential**.**

Example: Dropping mercury electrode

This electrode was first introduced by the Barker. The basic principle involved in this electrode is to control mercury flow through the capillary tube which is closed by the needle valve

ADVANTAGES

- This electrode is applicable to +0.4 to −1.8 V.

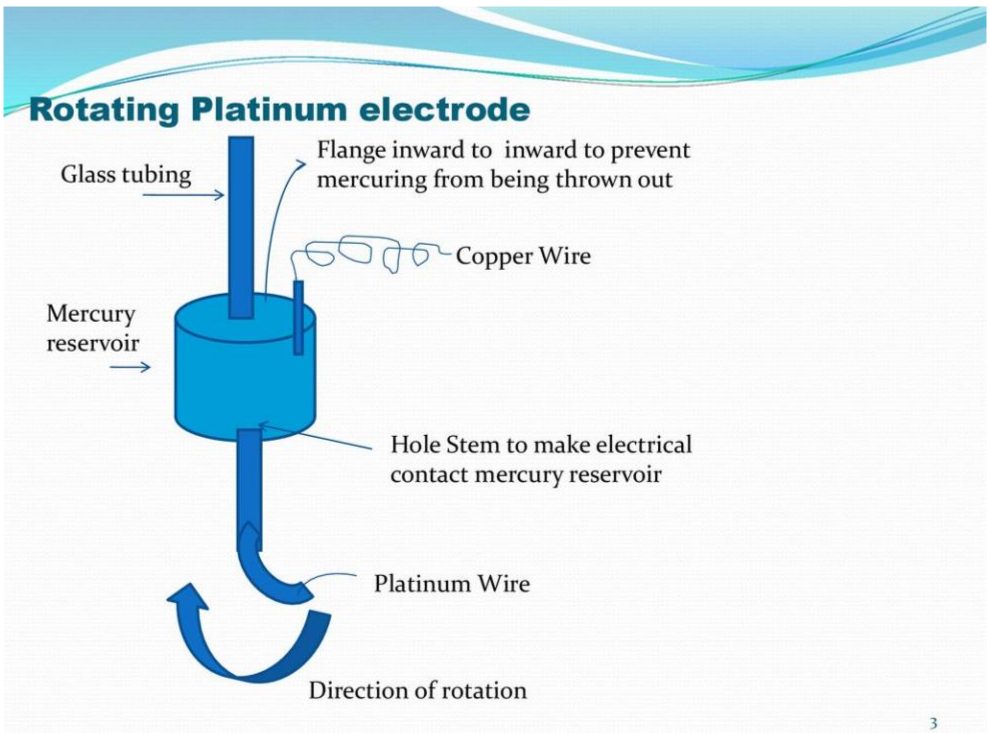
- Surface area is reproducible
- Constant renewal of electrode surface, poisoning effect can be removed
- Reduction of alkali metal ions can be obsevered due to large hydrogen overvoltage on Hg
- Surface area can be calculated by weght of drop
- Steady value is obtained by diffusion current

DISADVANTAGE:

- It can be oxidised easily hence avoided to used as anode
- capillary blocking

AUXILIARY ELECTRODE

It completes the circuit between the potentio state and the working electrode. Examples: Platinum electrode, Glass carbon electrode



REFERENCE ELECTRODE:

There are two types of reference electrodes

a) **External** – kept separated from solution through salt bridge or porous membrane
b) **Internal:** directly in contact with solution, preferred when high negative potential is required or salt bridge material affect adversely. It is made by coiling of around 15-20 cm of gauze silver wire into helix, coiling around DME. This electrode provides the

reference potential for the working electrode and for the auxiliary electrode. Examples: Silver–silver chloride electrode, Calomel electrode etc.

APPLICATIONS OF POLAROGRAPHY

- Used in the determination of the composition of the alloys.
- Used in the qualitative determination of the elements.
- Used in the estimation of the trace metals like Zn, Fe, Mn and Cu.
- Used in the determination of the free sulfur in petroleum fractions.
- Used in the determination of the vitamin C in the food beverages.
- Used in the functional group analysis.
- Used in the determination of the complex compositions.
- Used in the determination of the dissolved oxygen in the gases.
- Used in the determination of the local anesthetics (dyclonine)

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